Edge Computing in Industrial Communication

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***Abstract*—Edge computing processes data near where it is created, rather than sending it all to faraway cloud servers. Devices, gateways, or local servers handle data close to users or sensors. This reduces delays, saves bandwidth, improves privacy, and supports real-time applications. Edge computing is especially useful for the Internet of Things, smart cities, self-driving cars, healthcare, and industrial automation. This paper covers the basics of edge computing, its structure, benefits, uses, challenges, and future prospects.**

***Index Terms*—Edge Computing, Cloud Computing, Internet of Things, IoT, Low Latency, Distributed Systems**

I. Introduction

The way we manage data has changed dramatically in the past decade. With billions of devices online, it is no longer practical to send all data to distant servers. Edge computing addresses this problem by bringing the processing power closer to the data source, such as factory machines, pipeline sensors or assembly line cameras [1]. Instead of sending everything to a cloud data center, the data is processed at a local gateway or edge server and only key results or summaries are sent to the cloud. This architecture of field devices at the bottom, edge nodes in the middle and cloud at the top is the main setup of modern edge computing [2]. The industry is time critical. Factories are impatient! Robotic arms, conveyor belts, sensors and controllers have to respond within milliseconds, as the slightest delay can cause defects, damage to equipment or safety risks [3]. Edge computing helps by processing data at the source, enabling systems to identify problems early, respond immediately, and function smoothly without relying on the internet [1]. Another advantage is resilience—if the cloud connection drops, the edge system can continue to operate on its own. This independence is not just useful for factories in remote or unstable network areas, it is essential [2].

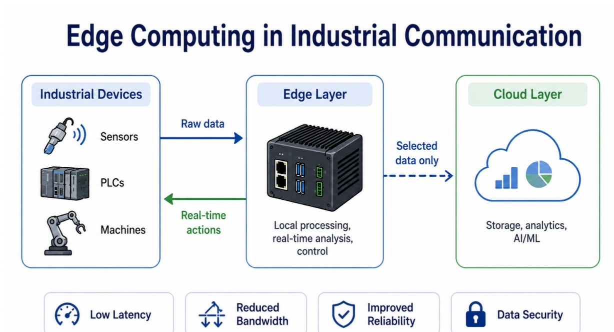

Fig. 1. Edge Computing in Industrial Communication

II. Basic Concept of Edge Computing

Every millisecond on a factory floor, sensors, PLCs, robots and cameras produce data that requires a rapid response. Sending all this data to a remote cloud server and waiting for a reply is too slow for industrial needs [1]. Edge computing solves this: a computing node is placed inside the industrial network, near the machines, so that data can be processed immediately without involving the cloud [2]. In the industrial

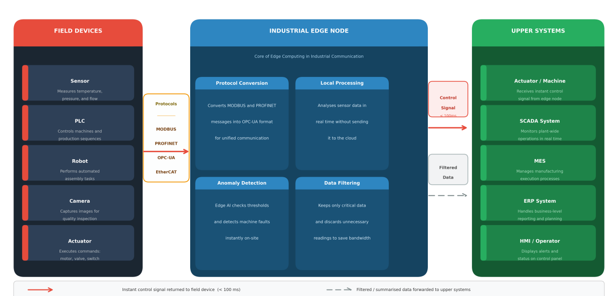

Fig. 2. Basic concept of edge computing

machines different communication protocols are used . For example a sensor could be using MODBUS and a robot OPC-UA. The edge node is a translator that translates all device communications to a single standard [6]. After the data is processed locally only important results like alarms, status updates etc. are sent to SCADA or ERP systems. All other data is on the factory floor [1,3].

If a machine fails, the edge node can detect the fault and send a shutdown command and notify the operator within milliseconds [3]. A cloud system cannot reply that fast, as sending data over the internet may take more than 500 ms [1]. In industry this speed is not optional, it is what makes production safe [2]

III. Architecture of Edge Computing

To understand edge computing in industry imagine a chain starting with physical machines at the bottom and ending with the remote cloud at the top. So, each link in that chain has a defined role, and the data is processed in the right place, at the right speed and with the right system [7]. As is shown in Fig. 2, This chain consists of four layers which work well together.

A. Cloud Layer

The top of the cloud is doing things that need not be done right now, such as long-term storage, retraining the model, ERP and MES reporting, and global monitoring The lower layers already filter the data as it comes to this layer [2,7]. The four layers are designed to work in concert, so that speed-sensitive tasks are kept close to the machines, while more complex analysis moves up the chain. This setup makes the system fast and scalable for industrial communication [1,7].

B. Fog Layer

The fog nodes are situated in between the edge and cloud. They collect data from edge gateways, short-term store and buffer it so that the cloud connection from being overloaded [3,7].

C. Edge Layer

It's got most of the smarts of the system in it. Data from machines is collected in raw form by edge gateways and local servers and processed locally, so that no data has to be sent to a remote server [2]. The edge layer also acts as translator between different industrial standards For example, an old MODBUS sensor and the new OPC-UA robot do not have direct communication, but an edge node link [6]. This layer can also be run live analytics with AI built in that detects anomalies and filters out non-essential data. only useful data is sent on [1]

D. Edge Gateway Server

The edge gateway serves as the gateway to the industrial edge network. It translates protocols such as MODBUS and OPC-UA, filters raw machine data and secures traffic between the factory floor and the outside network [6,7]

1) Cloudlet: A cloudlet is a small local computing unit deployed in proximity to industrial devices. It provides the kind of cloud-like processing power without the latency of a remote data.

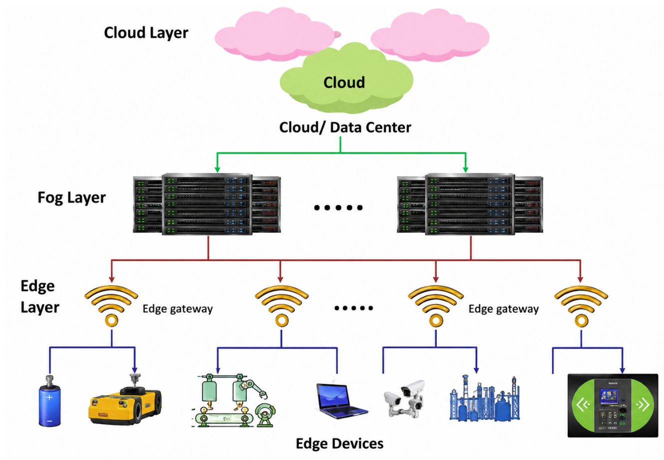

Fig. 3. Architecture of Edge Computing

2) Multi-Access Edge Computing (MEC): The ETSI standardises MEC and presents computing resources at the edge of the network. This enables industrial applications such as fault detection and real-time monitoring without a remote server [3, 7]. The main benefits are low latency, less network traffic, and easy scaling by adding more machines.

E. Components of Edge Computing

Edge computing includes several main parts that work together across the industrial network [7].

1) Edge Devices: These are the physical endpoints like sensors, cameras, PLCs and IoT modules that continuously collect raw data on the factory floor [1,7].

2) Edge Nodes: Between the field devices and the cloud are the cloud, routers and switches. They process data locally and decide what should go up the chain [7]

3) Edge Data Centers: Small data centers are deployed in close proximity to the industrial facility. They have local computing and storage so you don't need to rely on remote networks [7].

4) Edge Cloud: A distributed cloud architecture that is closer to the edge of the network.It provides cloud-like services to devices with no long latencies of a central data center [7].

5) Edge Analytics: Perform AI and machine learning modelling on edge nodes to detect faults and send alerts in real time without first sending data to the cloud [1,7].

6) Connectivity: All components are linked with fast and reliable communication links like industrial Ethernet, 5G or private wireless networks to enable data flow between layers [6,7].

IV. Difference Between Cloud Computing and Edge Computing

Cloud computing and edge computing are similar but not the same. Cloud Computing data centers to process and store the data in a centralised way. Computing Useful when lots of computing power and storage is needed. But it may cause delays since data need to go through the network[1,2].

With edge computing, certain data is processed near the source of the data. This reduces delay as well as network load. The key difference is where the data is processed. In cloud computing, processing happens in remote centralised servers. The edge computing, as processing takes place near the user or device[2,5].

TABLE I
COMPARISON BETWEEN CLOUD COMPUTING AND EDGE COMPUTING

Feature	**Cloud Computing**	**Edge Computing**
Processing location	Central cloud server	Near the device or user
Latency	Higher	Lower
Bandwidth use	More data sent to cloud	Less data sent to cloud
Storage capacity	Very high	Limited compared to cloud
Best use	Big data storage and analysis	Real-time applications

V. BENEFITS OF EDGE COMPUTING

Edge computing has several important advantages.

A. Low Latency

One of the key benefits of edge computing is low latency. Response times are quicker as data is processed nearer to where it is created. This is very important for real time systems such as autonomous vehicles, robotics and emergency health care etc.

B. Reduced Bandwidth Usage

Huge amount of data is generated by devices in the Internet of Things systems. All this data has to be sent to the cloud.This needs high band-width .

C. Improved Privacy and Security

Edge computing moves data processing and filtering closer to the source, reducing bandwidth requirements. Improved Privacy and Security Better Privacy Edge computing can enhance privacy by keeping sensitive data local to the device and not transmitting it to the cloud. For instance,in the healthcare systems, only the required data is sent to the cloud and the rest of the data can be processed locally.

D. Better Reliability

Edge computing can still operate when the connection to the cloud is poor or non-existent. This is helpful in industrial systems, remote areas and emergency

VI. APPLICATION AREAS OF EDGE COMPUTING

Edge computing is used in many modern fields.

A. Internet of Things

The Internet of Things devices are big data sources. Edge computing enables these systems to process data locally and react quickly. Examples are smart homes, smart factories and smart agriculture [1, 5].

B. Autonomous Vehicles

Autonomous vehicles must make decisions quickly. Cloud processing isn't always an option. Edge computing can process data from sensors and cameras in vehicles locally [2,3]

C. Healthcare

Edge computing can be used in the healthcare sector for patient monitoring, wearable devices, and emergency alert systems. It aids doctors and systems to react faster when the condition of a patient changes[5,7].

D. Industrial Automation

A factory has many machines, robots and sensors. Edge computing helps Industrial systems are used to control production processes, detect faults and monitor machines in real time [6,7].

E. Smart Cities

Smart cities use public safety systems, cameras and traffic sensors. Traffic management, congestion reduction and edge computing can analyse the data locally to improve public safety [1,7].

VII. LATENCY TEST BETWEEN EDGE AND CLOUD COMPUTING

To show the practical difference between edge computing and cloud computing,a simple latency test was performed. The IoT/client device used in this experiment was a mobile phone and the edge computing server was a laptop running Ubuntu 22.04. The cell phone sent sensor-like data to the laptop over a local Wi-Fi network.The laptop did the data locally, and replied. Latency was defined as the time from the request being sent until the response was received. This is because the processing is happening close to the user and not on a distant cloud server.The main idea of the experiment is edge computing.

A. Test Setup

The edge computing test used local communication between the mobile phone and the laptop edge server. The cloud computing test used an external internet server to compare the response time[9].

Fig. 4. Edge computing latency test diagram

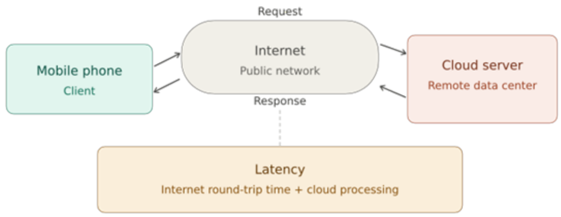

Fig. 5. Cloud computing latency test diagram

B. Edge Latency Test

A simple edge computing prototype was created using an Ubuntu 22.04 laptop and a mobile phone. The laptop worked as the edge server, and the mobile phone worked as the client device. Both devices were connected to the same home Wi-Fi network[9].

Mobile phone → Home Wi-Fi → Laptop edge server → Mobile phone

The laptop ran a Python Flask server. The mobile phone opened the server webpage using the laptop's local IP address:

http://192.168.178.78:5000

The terminal output showed that the mobile phone connected successfully:

```
192.168.178.117 - - "GET /
HTTP/1.1" 200 -

192.168.178.117 - - "POST /process
HTTP/1.1" 200 -
```

Here:

```
192.168.178.78 = Ubuntu laptop edge server
192.168.178.117 = Mobile phone client
```

The post/process request means the mobile phone sent data to the edge server. Status code 200 confirms successful communication.

C. Cloud Latency Test

For the cloud latency test, an HTTP request was sent to an online server through the internet. The communication path was:

Mobile phone → Internet → Cloud server → Internet → Mobile phone

The following command was used to measure the total request time:

```
curl -w "\nTotal time: %{time_total} seconds\n" \
-o /dev/null -s https://httpbin.org/get
```

The measured cloud result was:

```
Total time: 0.510079 seconds
```

This value can be converted into milliseconds as follows:

$$0.510079 \times 1000 = 510.079\,\mathrm{ms}$$

So the measured cloud http latency was 510ms. This is the total time it took for the http request to travel from the laptop to the remote internet server return and response [9]

VIII. Results and Discussion

The experiment shows the difference between local edge communication and cloud communication. In the edge test the mobile phone engaged with an adjacent laptop on a local Wi-Fi network. The request passed through the cloud test internet to a remote server and returned back.

The latency can be discussed using the following relation:

$$C = \frac{X}{L} + \frac{D}{R} \tag{1}$$

where C is latency, X is distance, L is network bandwidth, D is data rate, and R represents available computing resources[8].From the results, edge computing generally produced lower latency than cloud computing. This supports the main advantage of edge computing: processing data close to the source can reduce response time. The latency experiment in this paper is a simple way of measuring and comparing edge computing and cloud computing. Instead of estimates, the results are based on real measurements in two network environments: a home Wi-Fi network in Paderborn and the central Wi-Fi network at HSHL in Lippstadt. Both of these The setups show common scenarios found in industry and academia, a private network that is stable and a shared public network. They present the advantages and disadvantages of edge computing[9].

TABLE II
Measured Edge and Cloud Latency Results

Test Type	Test Location	Latency	Remarks
Edge Computing	Paderborn Home Wi-Fi	95 ms	Local processing near the device.
Edge Computing	Lippstadt HSHL Central Wi-Fi	422 ms	Network conditions affected latency.
Cloud Computing	Paderborn Home Wi-Fi	510 ms	Request travels through the internet.
Cloud Computing	Lippstadt HSHL Central Wi-Fi	2170 ms	Shared Wi-Fi increased delay.

IX. Comparative Analysis

Response Time: Edge vs. Cloud

The closest comparison is the Paderborn home wifi tests. The local network latency was 95 ms and cloud computing latency was 510 ms. This means that cloud processing took about 5.4 times longer than local edge processing of the same request-response cycle. The difference is baked into the design of each approach. Test the phone on the edge The phone talked over local Wi-Fi to a laptop close by, leaving the data on the local network. In the cloud test, the request went from the laptop to a remote server over the public internet and back, adding round trip internet time and server processing;

This finding supports the primary case for edge computing in industry, when you have to make decisions in milliseconds, such as stopping a machine, a cloud only system is much too slow to trigger a safety alarm or change a control setting. In this experiment, just sending data over the Internet took over 510 ms, much slower than real-time control systems require[7].

Impact of Network Quality

Influence of network quality The HSHL central WiFi tests show a more detailed result. Edge computing in hshl latency was as high as 422 ms, which is 4.4 times higher than the home Wi-Fi edge result. The latency for cloud computing at HSHL was even worse: 2170 ms. This shows that shared or crowded networks decelerate both methods but cloud computing is affected much more. Even with bad HSHL network, edge computing was 5.1 times faster than cloud computing in the same location (422 ms vs 2170 ms).

This result is of importance for industrial applications. Warehouses, factories and production places often have traffic from multiple devices on their networks at the same time. If an edge system uses this shared network for local communication, then the speed advantage of an edge system compared to the cloud is lowered. Nevertheless, the results demonstrate that edge computing is faster than cloud computing in each experiment. The key point is that you want to have a dedicated local network for edge communication, such as a private industrial Ethernet or a reserved wireless channel. important to realise the low-latency performance of edge computing [9].

Latency Formula and Its Implications

The latency relation used in the Results and Discussion section is:

$$C = \frac{X}{L} + \frac{D}{R} \tag{2}$$

In the home environment, the distance X between the mobile phone and the laptop edge server was small, the private Wi-Fi bandwidth L was dedicated and C was kept low. At HSHL, university's the common central Wi-Fi did a good job of limiting the bandwidth L available to any one device, so even though the physical distance X was similar, C went up. The same principle applies to cloud computing, but the distance X is the whole internet route to the remote server much larger than any local edge deployment. This distance is not helped by any amount of bandwidth improvement when strict millisecond response times are required[8].

TABLE III
EDGE VS. CLOUD: COMPARISON BASED ON EXPERIMENTAL RESULTS

Criterion	Edge Computing	Cloud Computing
Latency (stable network)	95 ms	510 ms
Latency (shared network)	422 ms	2170 ms
Speed advantage	Edge is 5.1–5.4× faster	
Network dependency	Local network	Internet
Offline operation	Yes	No
Suitable for real-time	Yes	Limited
Storage and analysis scale	Limited	High

The experimental results support that edge computing is the more suitable paradigm for industrial applications where time is of the essence.Cloud computing is still useful for large-scale storage, model training, long-term data analysis and enterprise reporting tasks. where response times of 500 ms or more are tolerable. measured latency differences for fault detection, process control, and safety response on the factory floor edge and cloud is too large to ignore. Hence, the results point to a hybrid architecture as the practical solution. The edge layer handles all latency sensitive local tasks while the cloud layer handles everything that does not require an immediate response[9].

X. IMPLEMENTATION IN INDUSTRIAL COMMUNICATION

A. Relation to Industrial Communication

The experiment was performed by a mobile phone, a Wi-Fi network and an Ubuntu laptop The same idea can be used in an industrial communication system. In the industrial environment the mobile phone can be seen as a sensor, PLC, robot, machine or an industrial IoT devices.The Ubuntu laptop works as an edge gateway, an industrial PC or a local server. The edge gateway instead of sending all the raw machine data to the cloud straight process data locally and only send out useful results, warnings or summaries [10].

TABLE IV
MAPPING THE PRACTICAL EXPERIMENT TO AN INDUSTRIAL EDGE COMPUTING SYSTEM

Experiment Component	Industrial Communication Equivalent
Mobile phone	Sensor, PLC, robot, machine, or IIoT device
Wi-Fi network	Industrial Ethernet, PROFINET, OPC UA, MQTT, 5G, or factory network
Ubuntu laptop	Edge gateway, industrial PC, or local server
Flask processing application	Local edge analytics or preprocessing software
Latency value	Communication response time in milliseconds

B. Real-World Examples in Industrial Communication

Real-world examples in industrial communication Edge computing is playing an important role in industrial communication as many processes need to take fast local decisions [10].Some real-world examples are:

1) **PCB Assembly and SMT Lines:** Camera inspection of component in PCB assembly placement, orientation and solder quality If a component is missing or misplaced, the edge system can instantly reject the board or halt the queue before additional bad boards are created.
2) **CNC Machining:** CNC machines are fitted with sensors that monitor vibration, temperature, and motor current. An edge node can detect localised tool wear or abnormal behaviour and alert the operator when a tool is going to break or damage the workpiece.
3) **Automotive Robotic Welding:** In the car manufacturing process, each weld is inspected by values such as

current, voltage and welding time. An edge system can quickly compare these values with quality limits and stop the robot if a defect is found or flag the part.

4) **Pharmaceutical Filling and Packaging:** Sensors in Pharmaceutical Manufacturing Cameras check the fill level, weight, label position and quality of the package. Edge computing can detect underfilled or contaminated products on the spot and remove them before final packaging. .

These examples illustrate that cloud-only processing is not suitable for time-critical work in industry This is also confirmed by the latency test in this paper,while cloud communication was 510 ms and even edge computing got 95 ms in stable local conditions 2170 ms over shared network. So, edge computing is more suitable for real-time monitoring, fault detection and quality control [9].

XI. Limitation and Challenges

Edge computing has many benefits, but it also has some challenges[7].

A. Security Issues

Edge devices are typically distributed across multiple sites. Which makes them harder to protect than cloud servers that are centralised. If an edge device is attacked, the entire system is can be impacted.

B. Limited Resources

Cloud edge devices usually have less computing power, memory and store than cloud Server-side applications need to be carefully designed.

C. Management Complexity

Managing too many edge devices is hard. Software updates, monitoring, maintenance and fault detection are complicated.

D. Data Consistency

Data is processed in different sites, which makes it difficult to maintain data consistency between the edge and cloud systems.

XII. Conclusion

Edge computing is a key technology for modern distributed systems. This brings computation closer to the data source and reduces the need for sending all data to remote cloud servers. It reduces latency, saves bandwidth, improves privacy and supports real time applications. Edge computing is a complement to the cloud, not an alternative to the cloud. The edge is used for fast local processing and the cloud is used for large-scale storage and analysis. The benefits of edge computing mean it will continue to be important in Internet of Things, smart cities, healthcare, industrial automation and autonomous systems.

XIII. Acknowledgment

The authors are grateful to Prof. Dr.-Ing. Ali Hayek for his valuable guidance, support and constructive feedback throughout this work. Under his guidance and leadership the quality and academic direction of the paper was elevated. The authors also attest that this paper is free from plagiarism and all sources outside of this paper have been cited. I used ChatGPT-5.5 only as a supporting tool for understanding the topic, improving clarity and generating illustrative images, and the final academic responsibility rests with the authors.

A. Team Member Contributions

All authors contributed to this work. Md Mostafizur Rahman participated in the testing, discussion and analytical part of the study. Turja Barua contributed to the architectures and comparison parts, contributing in the explanation of the system architecture and comparison of related topics. Deepak Kapil concentrated on applications and limitations, highlighting where the topic is valuable and where its issues still persist. In the concept and implementation part: The main idea and the practical application of it with the help of Moaz Chowdhury. All of these contributions allowed the paper to be more organised, intelligible and thorough.

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